

Surveillance Robot controlled using an Android app

Project Report

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Engineering

by

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This is to certify that the project entitled “*Surveillance Robot controlled using an Android app*” is a bonafide work of **Shaikh Shoeb Maroof Nasima** (12CO92), **Ansari Asgar Ali Shamshul Haque Shakina** (12CO106), **Khan Sufiyan Liyaqat Ali Kalimunnisa** (12CO81), **Mir Ibrahim Salim Farzana** (12CO82) submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of “**Bachelor of Engineering**” in **Department of Computer Engineering**.

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Project Approval for Bachelor of Engineering

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Declaration

I declare that this written submission represents my ideas in my own words and where others ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Title: Surveillance Robot controlled using an Android app

The robotics and automation industry which is ruled the sectors from manufacturing to household entertainments. It is widely used because of its simplicity and ability to modify to meet changes of needs. The project is designed to develop a robotic vehicle using android application for remote operation attached with wireless camera for monitoring purpose. The robot along with camera can wirelessly transmit real time video with night vision capabilities. This kind of robot can be helpful for spying purpose in war fields. The wifi technology is relatively new as compared to other technologies and there is huge potential of its growth and practical application. The android application loaded on mobile devices, can connect with security system and easy to use GUI. The security system then acts on these command and responds to the user. The CMOS camera and the motion detector are attached with security system for remote surveillance. A robot is a machine capable of carrying out a complex series of actions automatically, especially one programmable by a computer. A robot can be controlled by a human operator, sometimes from a great distance. In such type of applications wireless communication is more important. This paper also shows general idea and design of the robot. Surveillance security robot provides safety like man. Automatic patrolling vehicle for periodic patrolling in defined or a restricted area, the patrolling vehicle can move automatically to monitor the dead zones and capture the images by using the camera.

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Keywords And Glossary

Keywords :

Android, Robot, Bluetooth, Robotic control, Wi-Fi, Surveillance, Artificial Intelligence, Arduino

Glossary :

A

Artificial intelligence

Artificial intelligence (AI) is the intelligence exhibited by machines or software. It is an academic field of study which generally studies the goal of emulating human-like intelligence, though other variations of AI such as strong-AI and weak-AI are also studied. Major AI researchers and textbooks define this field as "the study and design of intelligent agents", where an intelligent agent is a system that perceives its environment and takes actions that maximize its chances of success.

Arduino

Arduino is an open-source computer hardware and software company, project and user community that designs and manufactures kits for building digital devices and interactive objects that can sense and control the physical world.[1] Arduino boards may be purchased preassembled, or as do-it-yourself kits; at the same time, the hardware design information is available for those who would like to assemble an Arduino from scratch.

ATMEGA 328

The ATmega328 is a single chip micro-controller created by Atmel and belongs to the megaAVR series. The Atmel 8-bit AVR RISC-based microcontroller combines 32 KB ISP flash memory with read-while-write capabilities, 1 KB EEPROM, 2 KB SRAM, 23 general purpose I/O lines, 32 general purpose working registers, three flexible timer/counters with compare modes, internal and external interrupts, serial programmable USART, a byte-oriented 2-wire serial interface, SPI serial port, 6-channel 10-bit A/D converter (8-channels in TQFP and QFN/MLF packages), programmable watchdog timer with internal oscillator, and five software selectable power saving modes. The device operates between 1.8-5.5 volts. The device achieves throughputs approaching 1 MIPS per Mhz.

B

BLUETOOTH

Bluetooth is a wireless technology standard for exchanging data over short distances (using short-wavelength UHF radio waves in the ISM band from 2.4 to 2.485 GHz[4]) from fixed and mobile devices, and building personal area networks (PANs). Invented by telecom vendor Ericsson in 1994,[5] it was originally conceived as a wireless alternative to RS-232 data cables. It can connect several devices, overcoming problems of synchronization.

C

CCTV

Closed-circuit television (CCTV), also known as video surveillance, is the use of video cameras to transmit a signal to a specific place, on a limited set of monitors. It differs from broadcast television in that the signal is not openly transmitted, though it may employ point to point (P2P), point to multipoint, or mesh wireless links. Though almost all video cameras fit this definition, the term is most often applied to those used for surveillance in areas that may need monitoring such as banks, casinos, airports, military installations, and convenience stores. Videotelephony is seldom called "CCTV" but the use of video in distance education, where it is an important tool, is often so called.

H

HC 05

These small size Bluetooth TTL transceiver modules are designed for serial communication (SPP - serial port profile). It allows your target device to both send or receive TTL data via Bluetooth technology without connecting a serial cable to your computer.

L

L293D

L293D is a dual H-bridge motor driver integrated circuit (IC). Motor drivers act as current amplifiers since they take a low-current control signal and provide a higher-current signal. This higher current signal is used to drive the motors.

R

Robot

A Robot is an automatic mechanical device often resembling a human or animal. Modern robots are usually an electro-mechanical machine guided by a computer program or electronic circuitry. Robots can be autonomous or semi-autonomous and range from humanoids such as Honda's Advanced Step in Innovative Mobility (ASIMO) and TOSY's TOSY Ping Pong Playing Robot (TOPIO) to industrial robots, collectively programmed swarm robots, and even microscopic nano robots.

S

Surveillance

Surveillance is the monitoring of the behavior, activities, or other changing information, usually of people for the purpose of influencing, managing, directing, or protecting them.

W

Wi-fi

Wi-Fi (or WiFi) is a local area wireless technology that allows an electronic device to participate in computer networking using 2.4 GHz UHF and 5 GHz SHF ISM radio bands.

The Wi-Fi Alliance defines Wi-Fi as any "wireless local area network" (WLAN) product based on the Institute of Electrical and Electronics Engineers' (IEEE) 802.11 standards".[1] However, the term "Wi-Fi" is used in general English as a synonym for "WLAN" since most modern WLANs are based on these standards. "Wi-Fi" is a trademark of the Wi-Fi Alliance. The "Wi-Fi CERTIFIED" trademark can only be used by Wi-Fi products that successfully complete Wi-Fi Alliance interoperability certification testing.

Chapter 1

Project Overview

1.1 Introduction

The advent of new high-speed technology and the growing computer Capacity provided realistic opportunity for new robot controls and realization of new methods of control theory. This technical improvement together with the need for high performance robots created faster, more accurate and more intelligent robots using new robots control devices, new drivers and advanced control algorithms.

This project describes a new economical solution of robot control systems .In general; the robots are controlled through wired network. The programming of the robot takes time if there is any change in the project the reprogramming has to be done. Thus they are not user friendly and worked along with the user preferences. To make a robot user-friendly and to get the multimedia tone in the control of the robot, they are designed to make user commanded work. The modern technology has to be implemented to do this.

For implementing the modern technology it should be known by all the users to make use of it. To reach and to full-fill all these needs we are using android mobile as a multimedia, user friendly device to control the robot. This idea is the motivation for this project and the main theme of the project. In this modern environment everybody uses smart phones which are a part of their day-to-day life. They use all their daily uses like newspaper reading, daily updates, social networking, and all the apps like home automation control, vehicle security, human body anatomy, health maintenance, etc has been designed in the form of applications which can be easily installed in their hand held smart phones. This project approached a robotic movement control through the smart phones.

Hence a dedicated application is created to control an embedded robotic hardware. The application controls the movement of the robot. The embedded hardware is developed on 8051 micro-controller and to be controlled by a Smart phone on the basis of Android platform. 8051 controller is to receive the AT commands from the Smart phone and takes the data and controls the motors of the robot by the motor driver L293D. The robot can able to move forward, reverse, left and right movements. The Smart phone is been interfaced to the device by using Bluetooth. A Bluetooth device HC-05 module is going to be added to 8051 micro-controller

to receive commands from smart phone. A wireless camera is mounted on the robot body for spying purpose even in complete darkness by using infrared lighting.

1.1.1 Motivation

It should be noted whenever customers want to buy any product through online from any e-commerce website he/she does visit the many e-commerce websites for getting the desire product. Like this customer surf lots of time in visiting of e-commerce websites for getting the desire product. The customer not only surfs lots of time in visiting of e-commerce sites, and quality of products, but also he/she suffers from limited option to choose the product.

The proposed architecture, by making use of location-based service, offers a solution to those problems. That is, for customers there is no need to visits many e-commerce websites for buying the desire the product. A user can buy desired product without visiting many e-commerce website as well as without wasting of time in visiting of e-commerce website. Location-Based Smart Shopping using Android provides a stage to the customer where the user can get information of a particular product available in stores nearby. Moreover, a user is also provided navigation facility which will direct him to the shop he wants to buy the product from

1.1.2 Advantages Over Current System

In various areas there is a need of constant surveillance. The current surveillance system includes monitoring by using CCTV cameras and other monitoring system. Mostly these systems are stationary and they can cover a limited area. These systems are mostly control manually or through a computer. They cannot be used to cover a larger area as well as they cannot be controlled using any mobile device. In short we can say that these systems arenâ™t dynamic enough much which gives the need for the development of a surveillance system which is more dynamic and can be controlled remotely.

This project is aimed at developing a surveillance system which can be controlled remotely by using an Android App. It includes a robot with a Wireless Camera attach to it. This robot captures the high resolution video feed and transmits it to the connected Android device which is used to control the robot.

1.2 Proposed System Architecture

Arduino mega 2560 board is used for the robot. A Bluetooth module is connected to the Arduino board. The android application establishes a connection with the Bluetooth module on the Arduino board.

Once connection is successfully established, the user can send controlling commands through the GUI of the android application. The command sent from the mobile application is received by the Bluetooth module which transmits it through serial communication to the Arduino board.

System architecture is shown in fig

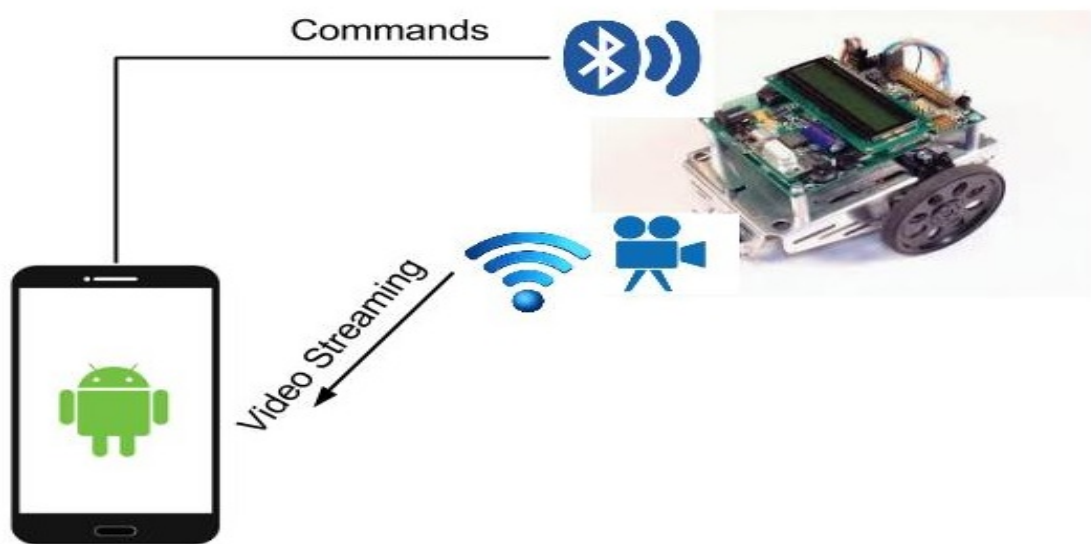


Figure 1.1: System Architecture

It consists of following blocks

- Blue-tooth module- HC-05
- Micro-Controller -ATMEGA 328
- DC motors
- Motor Driver H-Bridge
- 9v Dc battery‘

The robot in the project is made to move in all the four directions using the Android phone. The circuit is built around ATMEGA 328 controller board, Blue-tooth module HC-05, motor driver L293D (IC1), DC motors M1 and M2, and a few common components. The circuit uses two 9V batteries. First battery is used to power the ATMEGA 328 controller board and the other is used to power the motors. The regulated 5V supply for the rest of the circuit is provided by the ATMEGA 328 controller board itself. LED on the board indicates presence of power supply. Motor Driver H-Bridge is used to drive two motors which work on 9v DC batteries. DC

motors are interfaced to the Micro controller. The data received by the Blue-tooth module from Android smart phone is fed as input to the controller.

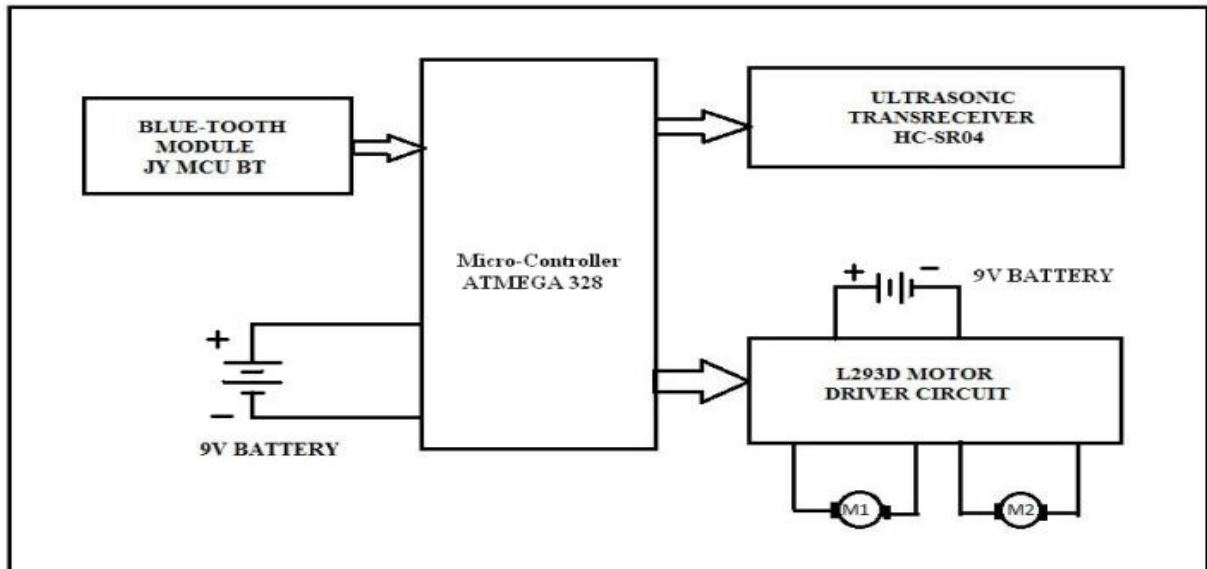


Figure 1.2: Block Diagram

1.2.1 Formulation of Problem With using Technology

1.2.1.1 Why Arduino?

There are many other microcontrollers and microcontroller platforms available for physical computing. Parallax Basic Stamp, Netmedia's BX-24, Phidgets, MIT's Handyboard, and many others offer similar functionality. All of these tools take the messy details of microcontroller programming and wrap it up in an easy-to-use package. Arduino also simplifies the process of working with microcontrollers, but it offers some advantage for teachers, students, and interested amateurs over other systems.

Inexpensive - Arduino boards are relatively inexpensive compared to other microcontroller platforms. The least expensive version of the Arduino module can be assembled by hand, and even the pre-assembled Arduino modules cost less than Cross-platform - The Arduino software runs on Windows, Macintosh OSX, and Linux operating systems. Most microcontroller systems are limited to Windows.

Simple, clear programming environment - The Arduino programming environment is easy-to-use for beginners, yet flexible enough for advanced users to take advantage of as well. For teachers, it's conveniently based on the Processing programming environment, so students learning to program in that environment will be familiar with the look and feel of Arduino.

Open source and extensible software- The Arduino software is published as open source tools, available for extension by experienced programmers. The language can be expanded through C++ libraries, and people wanting to understand the technical details can make the leap from Arduino to the AVR C programming language on which it's based. Similarly, you can add AVR-C code directly into your Arduino programs if you want to.

Open source and extensible hardware - The Arduino is based on Atmel's ATMEGA8 and ATMEGA168 microcontrollers. The plans for the modules are published under a Creative Commons license, so experienced circuit designers can make their own version of the module, extending it and improving it. Even relatively inexperienced users can build the breadboard version of the module in order to understand how it works and save money.

1.3 Organization of the Project

Chapter 2 contain Review of Literature. in this chapter we have studied and reviewed the previous work done on the topics related to our project. We did a detailed study on five such papers published in international journals. We have explained the pros, cons and the way we overcome the limitation of that project.

Chapter 3 contains Requirement analysis. In this chapter we give the detailed information on all the requirements of our project. It contains all the software and hardware related requiremets.

Chapter 4 contains the Project design. In this chapter we explain the project diagrammatically. All the diagrams are included in this chapter along with the explanation.

Chapter 5 contains the Implentation details. In this chapter all the details about the implementation on the project is given. It includes the assumptions and dependencied and implementation methodology as well. It aslo includes the detailed analysis and the description of project.

Chapter 6 contains the Results and Discussion. In this chapter we give all the test cases and the related results and discussion related to in.

Chapter 7 contains the Project time line. It includes the detailed advancement of the project in the form of a matrix time line and timeline chart.

Chapter 8 contains the task distribution. It gives a detailed explanation of the distribution on the workload.

Chapter 9 contains the Conclusion and Future scope. In this chapter we give the conclusion of our project and the future scope of the project.

Chapter 2

Review Of Literature

2.1 What is Surveillance robot ?

Surveillance robot is the robot used for the surveillance purpose. The remote areas are watched using the surveillance robots.

2.2 Wireless Controlled Surveillance Robot

2.2.1 Description

A mobile robot is a machine that is basically place or mounted on a movable platform and can be with the help of certain instructions. In today's world a lot of fields use mobile robots. Many of the complex robots that we now see have originated from the simpler mobile robots. This technology has increased many new applications in the industry.[4] The combination of mobile devices and robots are leading to new ideas in lots of fields.

The mobile devices are now being used in many of the industrial applications this is mainly because of the reason that they are portable and have a longer battery life as compared to a laptop. Also they have a data plan through a cell phone carrier which is convenient as we can interact with the mobile robot once the connection is established.

Mobile Robots: The mobile robots can be classified into different types. The track robot is the robot that uses tracks to move around. However such robots are costly to build.[10] Also they are not as flexible as the wheeled robots. The wheeled robots are the robots which use wheels for moving. Such robots can move only on smooth flat surfaces. The third type is the legged robots which are based on human form. They have legs which helps them to move around. These robots are very difficult to design.

Proposed System

The new age of technology such as Android, GSM has redefined communication. Most people nowadays have access to mobile phones and thus the world indeed has become a global village. At any given moment, any particular individual can be contacted with the mobile phone. New innovations and ideas can be generated from it that can further enhance its capabilities.

Technologies such as Infra-red, Bluetooth, Wi-Fi which has developed in recent years goes to show the very fact that improvements are in fact possible and these improvements have eased our life and the way we live.[11] Remote management of several home and office appliances is a subject of growing interest and in recent years we have seen many systems providing such controls.

Mobile robots are robots which have the ability to move around and interact with their environment and not just hinged to a particular place. There are many labs and research groups from various universities and industries which are completely dedicated on researching mobile robots, because of their immense potential and varied application in industry, military, security, and entertainment.

The robot is specially designed for surveillance purpose. The control mechanism is provided along with video transmission facility. The video transmission is practically achieved through high speed image transmission. Initially, the robot will be equipped with an Android smart-phone which will capture the scenario in front of it will transfer the images to the server on which the user will be controlling and watching the live feed.

2.2.2 Pros

1. Infrared LED: 8pcs infrared LED, automatic operate in dark environment Resolution: VGA (640x480)/ QVGA(320x240)/ QQVGA(160x120)
2. Motion detection to trigger alarm

2.2.3 Cons

1. Limited Frequency Range: The frequency range used for typical RF communication is near about 3KHz-3GHz. The use of channel separator increases the reliability but decreases the actual usable working frequency range.
2. Limited Functions: The limited number of channels causes less number of combinations and thus there are less numbers of available functions.

3. **Limited Working Range:** The working range of RF circuits with transmitters and receiver is very small. It starts from a few meters to a few kilometres. The working varies from circuits to circuits, but mainly depends on the values of physical components used in the circuit. Mainly Wi-Fi and Wi-Max wireless services are used in RF transmitter and receiver circuits. The following table shows the actual working range of different wireless standards that can be used in wireless communication.
4. **Reliability of Operation:** The RF circuits are very prone to errors due to external conditions such as EMI (Electro-Magnetic Interference), medium saturation, absorption due to repetitive reflections from surface. Hence the output recovered is not always what is expected. This might be a serious problem when working with scientific experimental components.
5. **Security reasons:** This is the main disadvantage of using a RF circuit and the main reason why RF circuits are not preferred today. The RF frequency band is available for almost all the users for data communication. So there might be a scenario where more than one user is trying to accommodate channel for its own communication. In such case the frequency band may get interference from another user. Or worst case would be, some user intentionally trying to jam our communication network. The RF jammer circuits are very easy to design; hence the question of security arises when RF circuit is used in the circuit. This security loop hole can be very dangerous when the robot is being used for very confidential purposes. In areas of military these security threats can produce disastrous outcomes.

2.2.4 How we overcome Those problem in Project

1. Wifi connection is used for the operation of robot.
2. This gives the high security.
3. It also provide much more reliability of operation working range it also includes wifi.
4. N type wifi is used for higher security purpose as well as for better range.

2.3 Surveillance Security Robot With Automatic Patrolling Vehicle

2.3.1 Description

The word surveillance may be applied to observation from a distance by means of electronic equipment (such as CCTV cameras), or interception of electronically transmitted information (such as Internet traffic or phone calls). It may also refer to simple, relatively no- or low-technology methods such as human intelligence agents and postal interception.

Surveillance is very useful to governments and law enforcement to maintain social control, recognize and monitor threats, and prevent/investigate criminal activity. However, many civil

rights and privacy groups, such as the Electronic Frontier Foundation and American Civil Liberties Union, have expressed concern that by allowing continual increases in government surveillance of citizens we will end up in a mass surveillance society, with extremely limited, or non-existent political and/or personal freedoms.[7] An automatic patrolling vehicle acts as a security patroller in the security system, which can monitor those dead zones of the traditional fixed surveillance system. The remote monitoring capabilities can also be enhanced by using the wireless network.[8] And the face detection system is adapted to record and analyze the invaders.

System Architecture

The proposed self-propelled monitoring and surveillance Vehicle can be divided into the following parts: Wireless IPCAM video capture system, face detection system, remote monitor and alarm transmitter system, RFID position detection systems, and cell phone monitoring and control system. The diagram of system architecture shows the self-propelled vehicle uses RFID technology to control the moving direction. An RFID tag is installed in the right hand side of the self-propelled vehicle. When the self-propelled vehicle moves to a predefined routing path installed with RFID reader, the RFID reader would detect the RFID tag and send the signals back to the server to show the detected position on the map to indicate the status of the automatic vehicle.

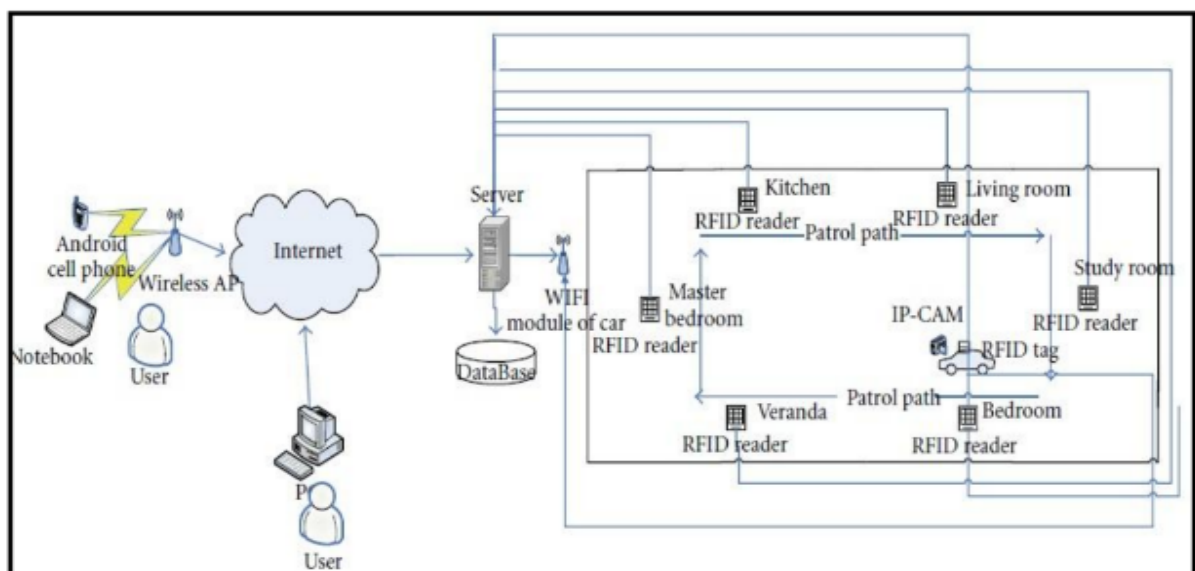


Figure 2.1: System Architecture

2.3.2 Pros

1. Image Sensor: 1/4" CMOS
2. High image video quality, two-way audio monitoring

3. Infrared LED: 8pcs infrared LED, automatic operate in dark environment
4. Resolution: VGA (640x480)/ QVGA(320x240)/ QQVGA(160x120)
5. Compression format: H.264
6. Allow remote Pan/Tilt control: 350° in pan, 60° in tilte
7. Motion detection to trigger alarm
8. KDM-6706AL is wireless model, Wi-Fi compliant with wireless standard IEEE 802.11b/g
9. Light weight
10. Omni directional coverage
11. High resolution images
12. Active participation in communication

2.3.3 Cons

1. Limited for only single stair
2. Limited angle detection (not able to detect faces under critical angle)
3. System itself having some limitations of capturing

2.3.4 How we overcome Those problem in Project

1. Movement is not limited or restricted as we used arduino robot
2. High quality camera is used for better capturing of videos
3. Light weight compression for edge point 264 is used for better compression

2.4 Android Based Security And Remote Surveillance System

2.4.1 Description

Now a day mobile devices are integrated with our everyday life. The security and remote surveillance system is increasingly prominent feature on the mobile phone. The modern home is integrated with many automation technologies. The user can control door lock, light, air conditioner and other devices using remote control. According to Khan, The Access control system used to allow only authorized members while the user away from their house.[12] When the system gets wrong password in three times than it signals to the door alarm. But this technology is very effective when using internet capable mobile devices.

Developments in cloud computing and mobile technology allow internet communication in automation and security systems to improve flexible and fast communication, such as Yale™s Locks Hardware new device.[13] This project exploits Near Field Communication through wifi

direct. Using X10 technology, the mobile device can control home security system. Priyare and tazil. Developed the home automation system through Bluetooth remote control. Sarijari et al. Implemented smart home system through zigbee communication. This system relays data and command via SMS message. Noor Azah Samsudin. Developed the food ordering system by using the wifi communication to order items with real time feedback. Saliyah Kahar presents comparative study of different wireless technology usage for mobile robot controller such as Bluetooth, WiFi or Wireless LAN and 3G. Saliyah Kahar, Riza Sulaiman .implements the mobile controlled robot, communicates through 3G technology to use advantages of multimedia features and internet speed. The 3G technology offers fast communication than 2G, is used for efficient multimedia data transmission. For long term communication, the 3G communication participate in important role.

The 3G can access high speed data rate at 2mbps. As in , applications created include performance-based wireless web, email, as well as video conferencing and multimedia services that blend voice and data streams. This paper discusses the development of security and video surveillance system, which communicates via wifi direct protocol. Android is currently leaded on mobile market share. This proposed system allows user to lock, unlock a door within short range only. The user can also monitor the house. The attached motion detector and CMOS camera is used for remote surveillance.

The mobile application requires password to increase the security of the system. The hardware on the door contains the AVR ATmega16 microcontroller to control a linear actuator for locking mechanism and to provide a link between camera and Android mobile. The wifi direct protocol was chosen as communication protocol because it is advanced wifi protocol, operate as adhoc network. The wifi direct protocol has large cover area, compared with Bluetooth. It can be used to communicate up to 200 meters away. The protocol incorporates data encryption for security and interference avoidance.

2.4.2 Pros

1. Omni directional coverage
2. Wi-Fi compliant with wireless standard IEEE 802.11b/g
3. Automatic operate in dark environment
4. Remote Pan/Tilt control: 350° in pan, 60° in tilte

2.4.3 Cons

1. Limited Working Range
2. Security reasons
3. Limited angle detection

4. Less reliability of Operation:

2.4.4 How we overcome Those problem in Project

1. Wifi connection is used for the operation of robot
2. his gives the high security
3. It also provide much more reliability of operation working range it also includes wifi
4. Movement is not limited or restricted as we used arduino robot
5. High quality camera is used for better capturing of videos

2.5 Smart Phone Controlled Robot Using ATMEGA328 Microcontroller

2.5.1 Description

Robots have been with us for less than 50 years but the idea of inanimate creations represents a sincere bid whose success is much older. But real robots did not come into existence until 1950s and 60s. With the growing invention of transistors and integrated circuits, computer industry added brains to the brawn of already existing machines.[22] In 1959, researchers illustrated the possibility of robotic manufacturing when they unveiled a computer-controlled milling machine. Bluetooth technology was created by telecom vendor Ericsson in 1994. A Bluetooth device is able to communicate with up to seven Bluetooth modules at same time through one link whose normal working area is within eight meters.

The basic function of Bluetooth serial module is replacing the serial port line by connecting one to Bluetooth master device and the other connecting to slave device. Today, robots are enjoying resurgence.[20] As computer processors are getting faster and inexpensive, robots can afford to get smarter. While, researchers are working on ways to help robots move and "think" more efficiently. Since most robots in use today are designed for specific tasks, our goal is to someday make universal robots that are flexible enough to do just about anything a human does and more.

Android is a software stack for mobile devices that includes an operating system and key applications. Android applications provide access to a wide range of useful libraries and tools that can be used to build rich applications.[21] Android also includes a full set of tools that provides developers with high productivity and deep insight into their applications. Bluetooth is a technology with an open standard specification for a radio frequency (RF)-based short range connectivity technology that changes the face of computing and wireless communication. The data received by the Bluetooth module from Android smart phone is fed as input to the controller. Thereby, the controller acts accordingly on the DC motors to move in the entire robot in all the four directions using the Android phone.

The circuit is built around an ATmega328 controller, ultrasonic transceiver module HC-SR04, Bluetooth module JYMCU BT, motor driver L293D (IC1), DC motors M1 and M2, and a few common components. The circuit uses two 9V batteries—one to power the ATmega328 controller and the other to power the motors. Regulated 5V supply for the rest of the circuit is provided by the ATmega328 controller itself. LED on the board indicates presence of power supply.

2.5.2 Pros

1. It can be used as a movable system
2. It can be controlled remotely
3. It does not require Man Power
4. It can be used for various critical application

2.5.3 Cons

1. No provision for video streaming
2. No dedicated app
3. Control using already developed application

2.5.4 How we overcome Those problem in Project

1. We have developed our own app to control the robot wirelessly
2. We use a network camera, mounted on the robot, for providing the live video feed
3. Our app includes live video streaming
4. It can be used for various real world applications

2.6 Smart Surveillance Monitoring System Using Raspberry PI and PIR Sensor

2.6.1 Description

Raspberry pi is a credit- card sized computer .It functions almost as a computer . There are various surveillance systems such as camera ,CCTV etc., [25] In these types of surveillance systems, the person who is stationary and is located in that particular area can only view what is happening in that place. Whereas, here ,even if the user is moving from one place to another

,he/she can keep track of what is happening in that particular place. Also another advantage is that it offers privacy on both sides since it is being viewed by only one person. The other major advantage is that it is a simple circuit .the operating system used here is Raspbian OS.Raspbian OS has to be installed so that the image can be transmitted to the smartphone.

Functional Description

The functions of the various components are given below: A. USB Camera: USB Camera captures the image and sends it to the USB port of the functionalities of the components are given below The various components of Raspberry- Pi are functionalities of the components are given below The various components of Raspberry- Pi are[26]

- SD Card Slot is used to install OS/booting/long term storage .The total memory of the SD card is about8GB.
- Micro USB Power Port provides 700mA at 5A
- RCA Video Out is connected to display if HDMI output is not used. It is mainly used to carry audio and video signals. They are otherwise called as A/V jacks.
- Audio Out Digital audio is obtained if HDMI is used to obtain stereo audio. Here analogue RCA connection is used.
- Ethernet Port is used to connect to the Internet. It also plays a role in updating, getting new software easier.
- HDMI OUT(High Definition Multimedia Interface) is used with HDTVs and monitors with HDMI input. Also HDMI-HDMI is used here.
- BROADCOM BCM 2835: It is otherwise defined as System on chip .It is a 700 MHz Processor. It has a Video core IV GPU.
- GPIO allows us to control and interact with real world . Raspberry Pi board. The camera model used here is USB Camera model 2.0.

2.6.2 Pros

1. Fast processing
2. High range
3. Can perform various task

2.6.3 Cons

1. There is no dedicated app
2. There is no provision for video streaming
3. Control using already developed application
4. It is very expensive
5. Even the camera used, is very expensive

2.6.4 How we overcome Those problem in Project

1. We use a network camera, mounted on the robot, for providing the live video feed
2. We have developed our own app to control the robot wirelessly
3. Our app includes live video streaming
4. It can be used for various real world applications
5. We use Arduino board which is cheaper than Raspberry pi
6. We use Android phone instead of expensive camera

2.7 Technological Review

2.7.1 Android (operating system)

Android is a mobile operating system (OS) based on the Linux kernel and currently developed by Google. With a user interface based on direct manipulation, Android is designed primarily for touchscreen mobile devices such as smartphones and tablet computers, with specialized user interfaces for televisions (Android TV), cars (Android Auto), and wrist watches (Android Wear). The OS uses touch inputs that loosely correspond to real-world actions, like swiping, tapping, pinching, and reverse pinching to manipulate on-screen objects, and a virtual keyboard. Despite being primarily designed for touch screen input, it also has been used in game consoles, digital cameras, regular PCs (e.g. the HP Slate 21) and other electronics.

As of July 2013, the Google Play store has had over one million Android applications ("apps") published, and over 50 billion applications downloaded. A developer survey conducted in April–May 2013 found that 71 % of mobile developers develop for Android. At Google I/O 2014, the company revealed that there were over one billion active monthly Android users, up from 538 million in June 2013. As of 2015, Android has the largest installed base of all general-purpose operating systems.

Android's source code is released by Google under open source licenses, although most Android devices ultimately ship with a combination of open source and proprietary software, including proprietary software developed and licensed by Google. Initially developed by Android, Inc., which Google backed financially and later bought in 2005, Android was unveiled in 2007, along with the founding of the Open Handset Alliance a consortium of hardware, software, and telecommunication companies devoted to advancing open standards for mobile devices.

2.7.2 Arduino Board

2.7.2.1 What is Arduino?

Arduino is a tool for making computers that can sense and control more of the physical world than your desktop computer. It's an open-source physical computing platform based on a simple microcontroller board, and a development environment for writing software for the board.

Arduino can be used to develop interactive objects, taking inputs from a variety of switches or sensors, and controlling a variety of lights, motors, and other physical outputs. Arduino projects can be stand-alone, or they can communicate with software running on your computer (e.g. Flash, Processing, MaxMSP.) The boards can be assembled by hand or purchased pre-assembled; the open-source IDE can be downloaded for free.

The Arduino programming language is an implementation of Wiring, a similar physical computing platform, which is based on the Processing multimedia programming environment.

2.7.2.2 Why Arduino?

There are many other microcontrollers and microcontroller platforms available for physical computing. Parallax Basic Stamp, Netmedia's BX-24, Phidgets, MIT's Handyboard, and many others offer similar functionality. All of these tools take the messy details of microcontroller programming and wrap it up in an easy-to-use package. Arduino also simplifies the process of working with microcontrollers, but it offers some advantage for teachers, students, and interested amateurs over other systems:

- Inexpensive - Arduino boards are relatively inexpensive compared to other microcontroller platforms. The least expensive version of the Arduino module can be assembled by hand, and even the pre-assembled Arduino modules cost less than \$50
- Cross-platform - The Arduino software runs on Windows, Macintosh OSX, and Linux operating systems. Most microcontroller systems are limited to Windows.
- Simple, clear programming environment - The Arduino programming environment is easy-to-use for beginners, yet flexible enough for advanced users to take advantage of as well. For teachers, it's conveniently based on the Processing programming environment, so students learning to program in that environment will be familiar with the look and feel of Arduino
- Open source and extensible software- The Arduino software is published as open source tools, available for extension by experienced programmers. The language can be expanded through C++ libraries, and people wanting to understand the technical details can make the leap from Arduino to the AVR C programming language on which it's based. Similarly, you can add AVR-C code directly into your Arduino programs if you want to.
- Open source and extensible hardware - The Arduino is based on Atmel's ATMEGA8

and ATMEGA168 microcontrollers. The plans for the modules are published under a Creative Commons license, so experienced circuit designers can make their own version of the module, extending it and improving it. Even relatively inexperienced users can build the breadboard version of the module in order to understand how it works and save money.

2.7.2.3 Bluetooth

Bluetooth is a wireless technology standard for exchanging data over short distances (using short-wavelength UHF radio waves in the ISM band from 2.4 to 2.485 GHz[4]) from fixed and mobile devices, and building personal area networks (PANs). Invented by telecom vendor Ericsson in 1994, it was originally conceived as a wireless alternative to RS-232 data cables. It can connect several devices, overcoming problems of synchronization.

Bluetooth is managed by the Bluetooth Special Interest Group (SIG), which has more than 25,000 member companies in the areas of telecommunication, computing, networking, and consumer electronics. The IEEE standardized Bluetooth as IEEE 802.15.1, but no longer maintains the standard. The Bluetooth SIG oversees development of the specification, manages the qualification program, and protects the trademarks. A manufacturer must make a device meet Bluetooth SIG standards to market it as a Bluetooth device. A network of patents apply to the technology, which are licensed to individual qualifying devices.

2.7.2.4 Wi-Fi

Wi-Fi (or WiFi) is a local area wireless technology that allows an electronic device to participate in computer networking using 2.4 GHz UHF and 5 GHz SHF ISM radio bands.

The Wi-Fi Alliance defines Wi-Fi as any "wireless local area network" (WLAN) product based on the Institute of Electrical and Electronics Engineers' (IEEE) 802.11 standards". However, the term "Wi-Fi" is used in general English as a synonym for "WLAN" since most modern WLANs are based on these standards. "Wi-Fi" is a trademark of the Wi-Fi Alliance. The "Wi-Fi CERTIFIED" trademark can only be used by Wi-Fi products that successfully complete Wi-Fi Alliance interoperability certification testing.

Many devices can use Wi-Fi, e.g. personal computers, video-game consoles, smartphones, digital cameras, tablet computers and digital audio players. These can connect to a network resource such as the Internet via a wireless network access point. Such an access point (or hotspot) has a range of about 20 meters (66 feet) indoors and a greater range outdoors. Hotspot coverage can comprise an area as small as a single room with walls that block radio waves, or as large as many square kilometres achieved by using multiple overlapping access points.

Depiction of a device sending information wirelessly to another device, both connected to the local network, in order to print a document.

Wi-Fi can be less secure than wired connections, such as Ethernet, because an intruder does not need a physical connection. Web pages that use TLS are secure, but unencrypted internet access can easily be detected by intruders. Because of this, Wi-Fi has adopted various encryption technologies. The early encryption WEP proved easy to break. Higher quality protocols (WPA, WPA2) were added later. An optional feature added in 2007, called Wi-Fi Protected Setup (WPS), had a serious flaw that allowed an attacker to recover the router's password. The Wi-Fi Alliance has since updated its test plan and certification program to ensure all newly certified devices resist attacks.

Chapter 3

Requirement Analysis

3.1 Platform Requirement :

3.1.1 Supportive Operating Systems for Server :

1. Ubuntu 12.4 and above The supported Operating Systems For server include Linux. Linux is used as server operating system.
2. Window 7 and above Microsoft Windows is a series of graphical interface operating systems developed, marketed, and sold by Microsoft. Microsoft introduced an operating environment named Windows on November 20, 1985 as a graphical operating system shell for MS-DOS in response to the growing interest in graphical user interfaces (GUIs).
3. Android OS Android is a mobile operating system (OS) based on the Linux kernel and currently developed by Google. With a user interface based on direct manipulation, Android is designed primarily for touchscreen mobile devices such as smartphones and tablet computers, with specialized user interfaces for televisions (Android TV), cars (Android Auto), and wrist watches (Android Wear).

3.1.2 Supportive Operating Systems for Client:

1. Android OS Android is open source operating system based on the Linux kernel and currently developed by Google. With a user interface based on direct manipulation

3.2 Software Requirement :

The Software Requirements in this project include:

3.2.1 Eclipse : Eclipse Luna (4.4.1)

Eclipse is an integrated development environment (IDE). It contains a base workspace and an extensible plug-in system for customizing the environment.

3.2.2 JDK : Java Platform (JDK) 8u25

The Java Development Kit (JDK) is an implementation of either one of the Java SE, Java EE or Java ME platforms released by Oracle Corporation in the form of a binary product aimed at Java developers on Solaris, Linux, Mac OS X or Windows. The JDK includes a private JVM and a few other resources to finish the recipe to a Java Application.

3.2.3 JVM : JVM Version 8

A Java virtual machine (JVM) is an abstract computing machine. There are three notions of the JVM: specification, implementation, and instance. The specification is a book that formally describes what is required of a JVM implementation. Having a single specification ensures all implementations are interoperable. A JVM implementation is a computer program that meets the requirements of the JVM specification in a compliant and preferably performant manner. An instance of the JVM is a process that executes a computer program compiled into Java bytecode.

3.3 Hardware Requirement :

The Hardware components required for our project are Min 1 GB of RAM,10 GB HDD,Dual core processor for the machine on which development will be done,Robot kit,IP Camera,Bluetooth Module etc for developing the robot.

3.3.1 Hardware Required For Project Development:

1. Arduino robot environment
2. Rotor motor
3. Batteries
4. Camera module
5. Wifi module
6. Chassis

Chapter 4

Project Design

4.1 Design Approach

4.1.1 Front End Designs

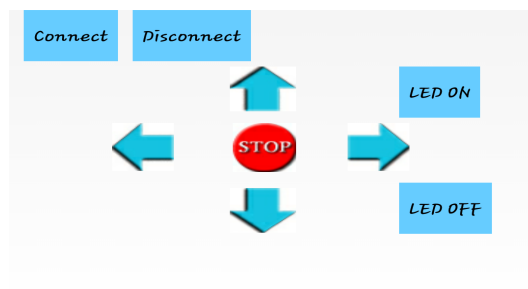
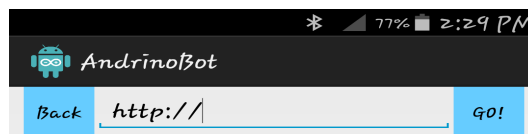


Figure 4.1: Front End

4.2 Software Architectural Designs

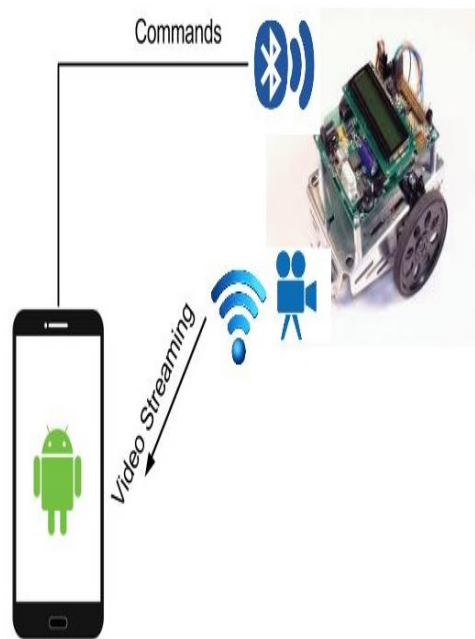


Figure 4.2: Software Architectural Designs

4.2.1 Component Diagram

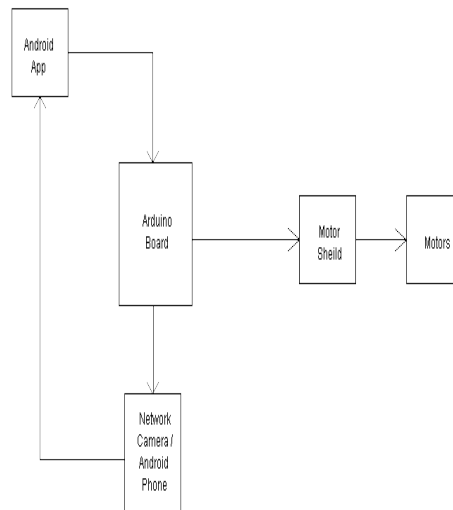


Figure 4.3: Component Diagram

4.2.2 Deployment Diagram

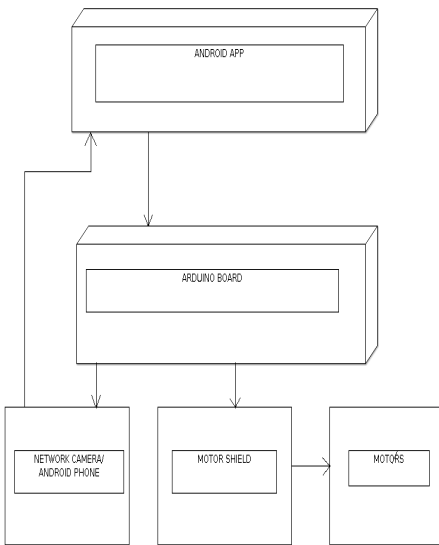


Figure 4.4: Deployment Diagram

4.3 Work-flow Design

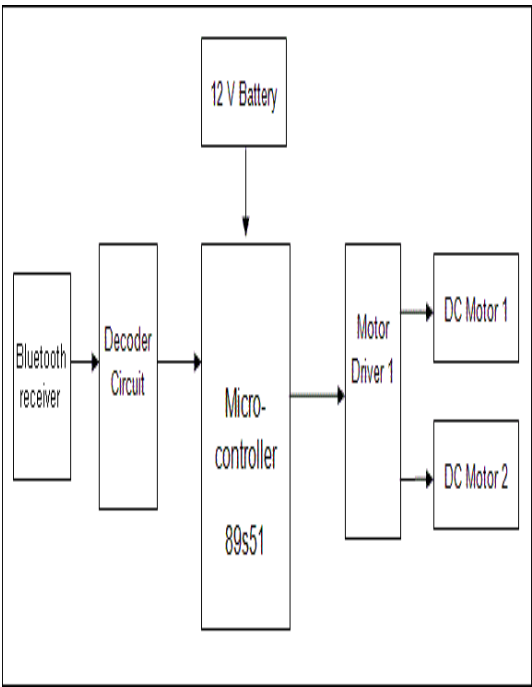


Figure 4.5: Work-Flow Design

4.4 Flow Graph

4.4.1 DFD LEVEL

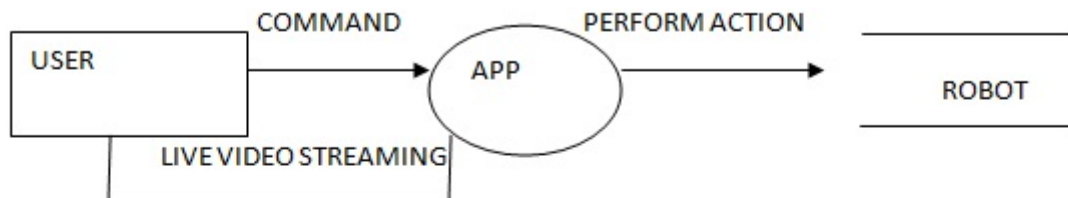


Figure 4.6: DFD Level 0

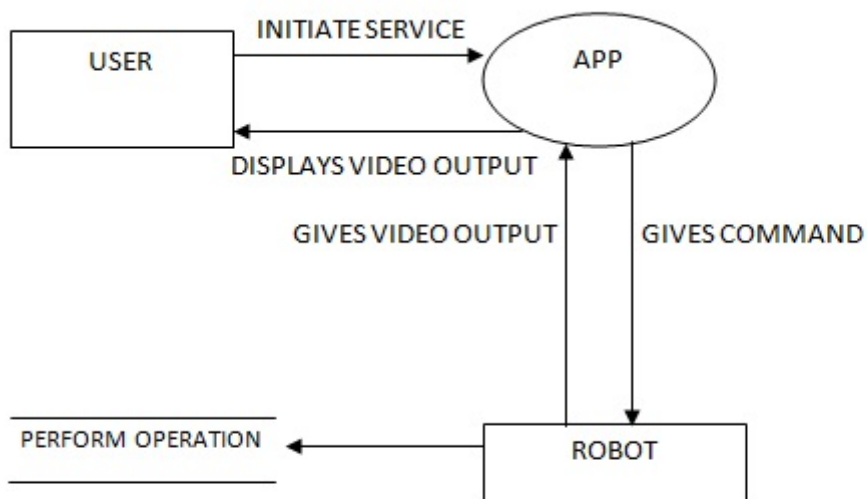


Figure 4.7: DFD Level 1

Chapter 5

Implementation Details

5.1 Assumptions And Dependencies

The new age of technology such as Android, GSM has redefined communication. Most people nowadays have access to mobile phones and thus the world indeed has become a global village. At any given moment, any particular individual can be contacted with the mobile phone. New innovations and ideas can be generated from it that can further enhance its capabilities. Technologies such as Infra-red, Bluetooth, Wi-Fi which has developed in recent years goes to show the very fact that improvements are in fact possible and these improvements have eased our life and the way we live. Remote management of several home and office appliances is a subject of growing interest and in recent years we have seen many systems providing such controls.

Mobile robots are robots which have the ability to move around and interact with their environment and not just hinged to a particular place. There are many labs and research groups from various universities and industries which are completely dedicated on researching mobile robots, because of their immense potential and varied application in industry, military, security, and entertainment.

The robot is specially designed for surveillance purpose. The control mechanism is provided along with video transmission facility. The video transmission is practically achieved through high speed image transmission. Initially, the robot will be equipped with an Android Smartphone which will capture the scenario in front of it will transfer the images to the server on which the user will be controlling and watching the live feed.

5.2 Implementation Methodologies

The project is designed to control a robotic vehicle using an android application. Bluetooth device is interfaced to the control unit on the robot for sensing the signals transmitted by the android application. This data is conveyed to the control unit which moves the robot as desired.

An Atmel 89C51 microcontroller is used in this project as control device.

Remote operation is achieved by any smart-phone/Tablet etc., with Android OS, upon a GUI (Graphical User Interface) based touch screen operation. Transmitting end uses an android application device remote through which commands are transmitted. At the receiver end, these commands are used for controlling the robot in all directions such as forward, backward and left or right and captures the video and transmits to TV through RF signal. At the receiving end the movement is achieved by two motors that are interfaced to the microcontroller.

Serial communication data sent from the android application is received by a Bluetooth receiver interfaced to the microcontroller. The program on the microcontroller refers to the serial data to generate respective output based on the input data to operate the motors through a motor driver IC. The motors are interfaced to the control unit through motor driver IC.

5.2.1 Modular Description of Project

5.2.1.1 Bluetooth module

The module provides a method to connect wirelessly with a PC or Bluetooth phone to transmit/receive embedded data such as GPS data, ADC voltage reading and other parameters. Bluetooth module JY MCU BT used in the project can be connected to any device, via built in UART interface to communicate with other Bluetooth-enabled devices such as mobile phones, handheld computers and laptops. The module runs on a 3.6V to 6V supply.

5.2.1.2 Arduino

Arduino micro-controller is intended for artists, designers, hobbyists and anyone interested in creating interactive objects or environments.

5.2.1.3 L293d

L293D is a dual H-bridge motor driver integrated circuit (IC). Motor drivers act as current amplifiers since they take a low-current control signal and provide a higher-current signal to run four solenoids, two DC motors or one bi-polar or unipolar stepper with up to 600 mA per channel using the L293D. These are known as the drivers in the Ada fruit Motor shield.

The L293 and L293D are quadruple high-current half-H drivers. In L293 is designed to provide bidirectional drive currents up to 1 A at voltage range from 4.5 V to 36 V. The L293D is designed to provide bidirectional drive currents of up to 600-mA at voltage range from 4.5 V to 36 V. Both the devices are designed to drive inductive loads such as relays, dc and bipolar stepping motor as well as other high-current/high-voltage loads in positive-supply applications.

The Android application is basically divided into two modules i.e.

1. Video Streaming Module
2. Robot control module

A network camera or an Android mobile phone is mounted on the robot, which our application fetches the live video streaming display it. This video is achieved using WIFI technology.

The second module is the control module. Our application provides a GUI to control the robot wirelessly. This control is achieved using Bluetooth technology. Button are used to control the robot in forward, backward, left or right direction.

5.3 Detailed Analysis and Description of Project

The android application will be used to search for products and view stores at which those products are available. The android application will need to communicate to a GPS application within the mobile phone, which in turn communicates with a physical GPS device to find the location of the user. The GPS will provide the mobile application with locations of both the user and the stores and the distance between them, but it will also provide maps and the functionality to display the application's data on the map. The functionality provided by the GPS will be embedded into the application in order for the user to be able to use the functions in the application in a seamlessly manner.

Since this is a data-centric product it will need somewhere to store the data. For that, a database will be used. The android application will communicate with the database. The user will use the application to get data from the database while the vendor will also add and modify data, which will be monitored by the admin. All of the database communication will go over the Internet. The mobile application has some restrictions about the resource allocation.

5.3.1 Usecase Report

Title:	Surveillance Robot Controlled Using An Android App
Description:	In this project we can control the surveillance robot using an android app. This is done through the help of a user-friendly GUI.
Primary Actor:	The one who uses the mobile
Pre-conditions:	Installation of Android app on our mobile
Post-conditions:	controlled a robot using app .
Main Success Scenario:	1. Featching a live videostreaming on android app. 2. video is taken from a mobile which is mounted on a robot and then send to our android app. 3. The Procedures starts by seeing a video then we can control the robot left,right,forward,bavckward easily.
Frequency of Use:	User can use many times
System Requirement:	android app.

Table 5.1: Usecase Report

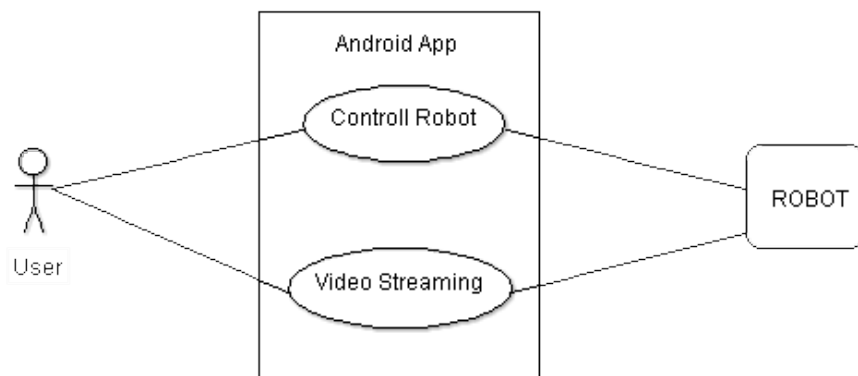


Figure 5.1: Use Case Diagram

5.4 Class Diagram

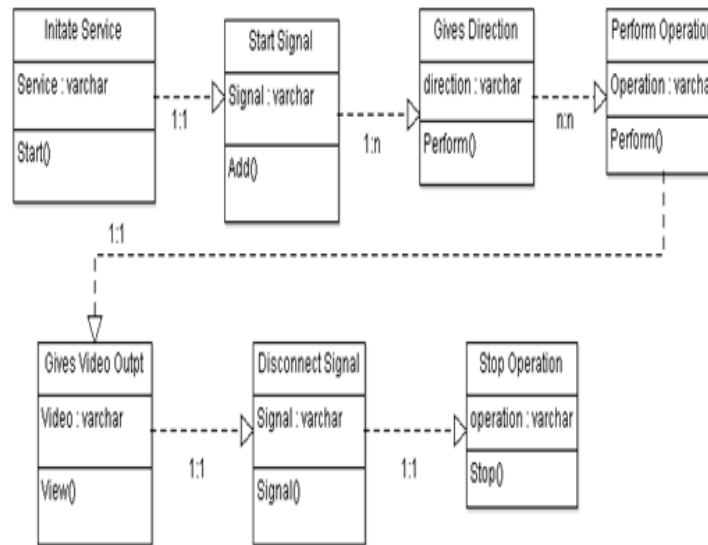


Figure 5.2: Class Diagram

5.5 Class Report

Title:	Surveillance Robot Controlled Using An Android App
Description:	In this project we can control the surveillance robot using an android app. This is done through the help of a user-friendly GUI. We can fetch a video using two mobiles one is mounted on a robot and another one is at a user hand.
Primary Actor:	The one who uses the Android App
Preconditions:	User connect a robot using bluetooth
Post conditions:	watching a live video streaming and controlling it
Comparative Website :	1. Data Independence 2. Controlled Redundancy 3. Accuracy 4. Security 5. Performance
Arduino:	Arduino is a tool for making computers that can sense and control more of the physical world than your desktop computers.

Table 5.2: Class Diagram Report

Chapter 6

Results and Discussion

6.1 Test Cases

1. When i have tested my robot i have found that the range of bluetooth is around 10 meter.
2. In dark situation there was no light so that i used LED flash light for better visibility.
3. Our robot is easily move on the rough area.

6.2 Results Discussion

We have two results i.e. the hardware and the software result. The hardware includes the robot which runs on DC motors. The input to the motors is provided by the L293D motor driver shield. The input to the driver shield is provided by the arduino board.

The navigational inputs are given by the user to the arduino board using the android application via Bluetooth. The arduino board, on receiving the signal, processes it and produces the appropriate output. The communication between the android application and the arduino board takes place using the Bluetooth module which is interfaced with the arduino board. It provides serial communication between the application and the Arduino.

Chapter 7

Project Time Line

7.1 Project Time Line Matrix

		Name	Duration	Start	Finish	Predecessors
1		1(a) Requirements Gathering	8 days	1/1/15 8:00 AM	12/1/15 5:00 PM	
2		1(b) Confirm Requirements	1 day	13/1/15 8:00 ...	13/1/15 5:00 PM	
3		2(a) Frontend-User Interface	7 days	14/1/15 8:00 ...	22/1/15 5:00 PM	
4		2(b) Backend Robot Designing	10 days	23/1/15 8:00 ...	5/2/15 5:00 PM	
5		3(a) Frontend Coding	12 days	6/2/15 8:00 AM	23/2/15 5:00 PM	
6		3(b) Robot Creation	10 days	24/2/15 8:00 ...	9/3/15 5:00 PM	
7		3(c) Coding for GUI	8 days	10/3/15 8:00 ...	19/3/15 5:00 PM	
8		3(d) Creation of test case	9 days	20/3/15 8:00 ...	1/4/15 5:00 PM	
9		4(a) Unit Test	6 days	2/4/15 8:00 AM	9/4/15 5:00 PM	
10		4(b) System Test	7 days	10/4/15 8:00 ...	20/4/15 5:00 PM	
11		4(c) Functional Test	6 days	21/4/15 8:00 ...	28/4/15 5:00 PM	
12		5(a) Deployment	2 days	29/4/15 8:00 ...	30/4/15 5:00 PM	

Figure 7.1: Time Line Matrix



Chapter 8

Task Distribution

8.1 Distribution of Workload

8.1.1 Scheduled Working Activities

Activity	Time Period	Comment
Requirement Gathering	08 Days	Requirement gathering has took placed through searching on in-ternet and taking the ideas, shar-ing the views among group members.
Planning	04 Days	Planing has done by Reviewing of literature of IEEE papers and by taking the walkthrough.
Design	04 Days	Designing has done by creat-ing UML diagram, By creating Charts,
Implementation	90 Days	Implementation has done First creating the backend and then front end module by module.
Testing	10 Days	Testing has done by perfoming unit testing, alpha & Beta Test-ing, integrated testing and sys-tem testing.
Deployment	05 Days	Deployment has done by in-stalling project on the server.

Table 8.1: Scheduled Working Activities

8.1.2 Members activities or task

Member	Activity	Time Period	Start Date	End Date	Comment
M1, M2, M3, M4	Requirement Gathering	4 Days	01/01/15	04/01/15	M1 and M2 has performed the searching for project requirement on the internet by reviewing the related literature and by analysing the related project which is already available in the market. Regularly inform to the other member of team.
M1, M2, M3, M4	Analysing of the requirement	3 Days	05/01/15	07/01/15	M1, M2, M3, M4 done the requirement analysing of project by sharing the ideas, and by discussing on related information which is gathered by the M1, And M2. M3 and M4 has created the list of requirement after every meeting
M1, M2, M3, M4	Finalising the requirement	1 Day	08/01/15	08/01/15	Whole team finalise the requirement. M1 and M4 has created a list of finalise requirement.
M1, M2, M3, M4	Planning	4 Days	09/01/15	12/01/15	Planning has done by walkthrough and by analysing the available product. M2 and M3 creates a list of function which will be implemented in the project. Each and every module group were discussed in every group meeting and M1 and M2 creates a blue print for project .

M1, M2	Front End design	4 Days	13/01/15	16/01/15	M1 and M2 creates the UML diagram for front end of the system and data flow diagrams and informed to the whole team regularly.
M3, M4	Back End design	4 Days	13/01/15	16/01/15	M3 and M4 creates the UML diagram for back end of the system and data flow diagrams and informed to the whole team regularly.
M3, M4	Installation of tools and technology for front end	1 Days	17/01/15	17/01/15	M3 and M4 installed the all the require tools which is use for front end design.
M1, M2	Installation of tools and technology for back end	2 Days	17/01/15	19/01/15	M3 and M4 installed all the require tools which is use for back end design.
M3, M4	Implementation of GUI	4 Days	20/01/15	24/01/15	M3 and M4 creates the GUI of the project and informed to other member.
M1	Implementation of Bluetooth module	20 Days	20/01/15	10/02/15	M1 implemented the Bluetooth module and discuss on it with other team member
M2	Implementation of Camera Module	12 Days	20/01/15	02/02/15	M2 implemented the Camera Module and discuss on it with other team member

M3, M4	Bluetooth module with GUI	8 Days	07/02/15	14/02/15	M3 and M4 makes the conection with GUI and discuss on it with other team member regularly.
M1, M2	Component connectivity	4 Days	06/02/15	10/02/15	M1 and M2 created the Component connectivity with the Bluetooth and Camera Module.
M3, M4	GUI Connectivity	4 Days	11/02/15	14/02/15	M3 and M4 created the connectivity GUI
M3	Indexing	8 Days	26/02/15	04/03/15	M3 implements indexing programs in the Android. And Connect it with the GUI. He explain the code to other memeber of team.

M4	Connectivity of Android App with Robot	2 Days	05/03/15	07/03/15	M4 makes the indexing program connectivity with Robot. And informed to the other member of team.
M3,M4	Implementation of Android App codes	5 Days	07/03/15	12/03/15	M3 and M4 implemented Android App codes regularly discussed this implementation with the other member of team.
M1, M2, M3, M4	Integration of all modules	10 Days	16/03/15	25/03/15	M1, M2,M3 and M4 integrated all the module. Implemented whole system properly.
M1,M2	Unit testing	2 Days	26/03/15	28/03/15	M1 and M2 performed the unit testing and noted down results and discuss with other member of team.
M3, M4	Integrated testing	5 Days	26/03/15	30/03/15	M3 and M4 performed the integrated testing and noted down results and discuss the result of testing with other member.
M1, M2, M3, M4	Deployment	â	30/03/2015	â	â

Table 8.2: Member Activities and Task

Chapter 9

Conclusion and Future Scope

9.1 Conclusion

We have successfully implemented the working of the wireless video surveillance robot controlled using android mobile device. The robot is successfully controlled using the android application through the wireless Bluetooth technology. Even the real time video feed is successfully achieved using the Wi-Fi technology on our designed android application.

9.2 Future Scope

Surveillance is needed in almost every field. It could be a great solution to various problems or situation where wireless Surveillance is needed our project has tremendous scope as it uses the latest technology in the market.

Our application uses the android OS which is currently the most used OS and also has a great future scope. The Surveillance robot can be controlled remotely using the android application; this gives it a huge scope for future application.

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Chapter 10

Published Paper

1. Shaikh Shoeb, Mir Ibrahim, Ansari Asagr Ali, Khan Sufiyan, Kalpana Bodke. *Wireless Video Surveillance Robot Controlled Using Android Mobile Device*. International Conference On "Emerging Trends Of Technology, volume-Vivruti-2015
2. Shaikh Shoeb, Mir Ibrahim, Ansari Asagr Ali, Khan Sufiyan, Kalpana Bodke. *Wireless Video Surveillance Robot Controlled Using Android Mobile Device*. International Journal Of Advance Foundation And Research In Science And Engineering, Volume-1, Special Issue Vivruti-2015
3. Shaikh Shoeb, Mir Ibrahim, Ansari Asagr Ali, Khan Sufiyan, Kalpana Bodke. *Wireless Video Surveillance Robot Controlled Using Android Mobile Device*. International Journal Of Advance Research In Computer Science, Volume April-2015

Appendix I

10.1 Installation Of An Andrinobot Application

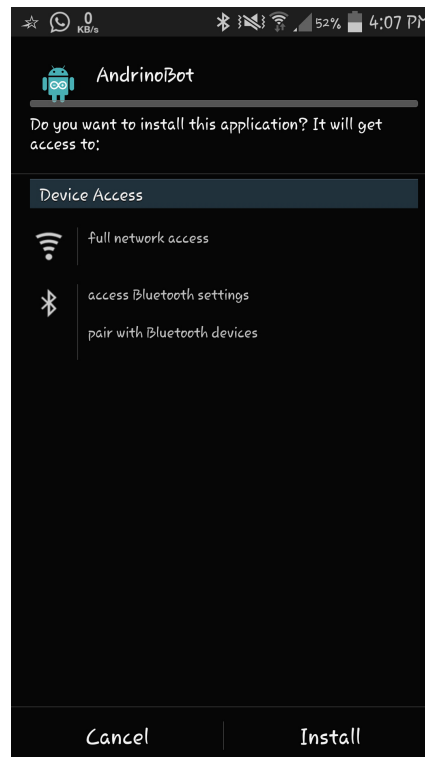


Figure 10.1: Installation Of An Andrinobot Application

10.2 Open The Android Application

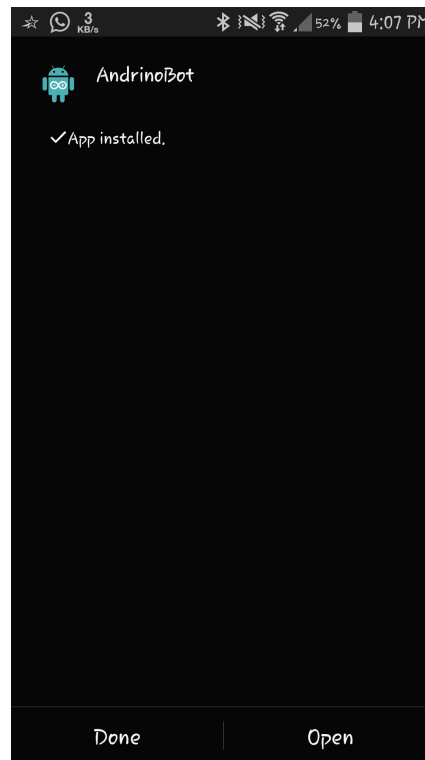


Figure 10.2: Open The Android Application

10.3 The Application Gets Started

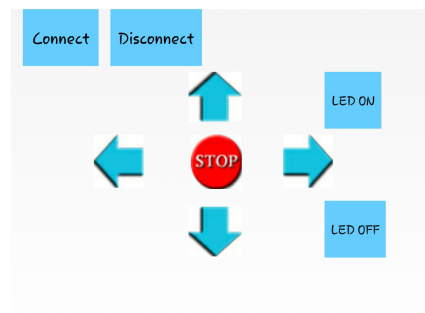
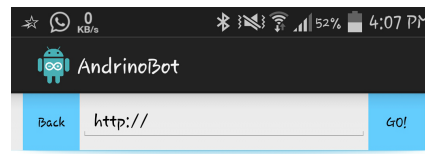


Figure 10.3: The Application Gets Started

10.4 Connect A Mobile Through wifi Direct



.png

Figure 10.4: Connect A Mobile Through wifi Direct

10.5. Now Connect A Mobile Through IP Address Of The Phone Which Is Mounted On A Robot

10.5 Now Connect A Mobile Through IP Address Of The Phone Which Is Mounted On A Robot

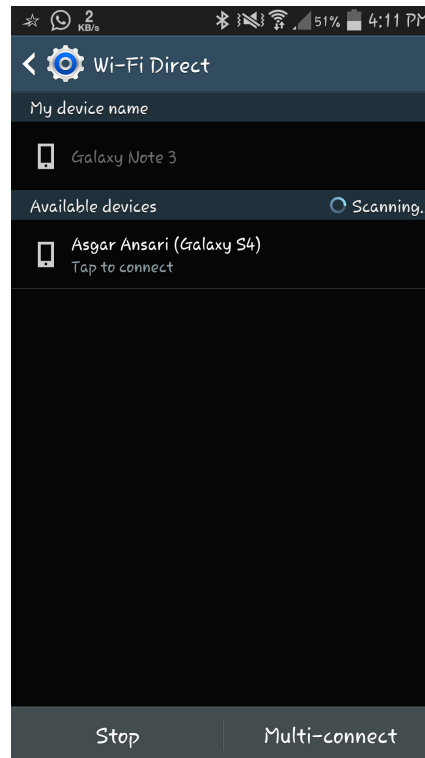


Figure 10.5: Now Connect A Mobile Through IP Address Of The Phone Which Is Mounted On A Robot

10.6 Once The Mobile Is Connect Watch The Live Video Streaming

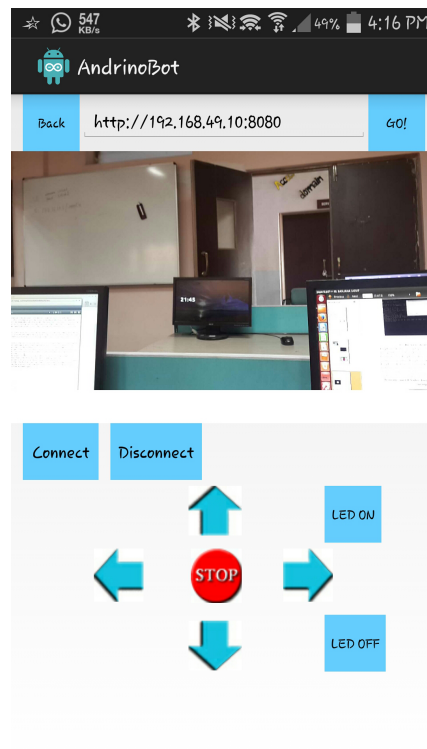


Figure 10.6: Once The Mobile Is Connect Watch The Live Video Streaming

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